

- 33** A piece of wire X has resistivity ρ , length L and cross-sectional area A . Wire X has a resistance R .

A second piece of wire Y is made of a different metal. It has the same resistance as X but has twice the length of X.

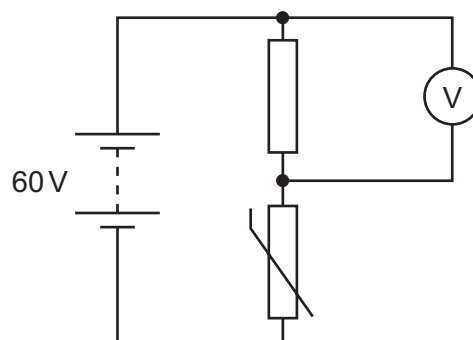
Which row gives possible values for the resistivity and the cross-sectional area of Y?

	resistivity	cross-sectional area
A	$\frac{1}{2}\rho$	$\frac{1}{2}A$
B	$\frac{1}{2}\rho$	A
C	ρ	$\frac{1}{2}A$
D	2ρ	A

- 34** A cell of electromotive force (e.m.f.) E delivers a charge Q to an external circuit.

Which statement is correct?

- A** The energy dissipated in the external circuit is EQ .
- B** The energy dissipated within the cell is EQ .
- C** The external resistance is EQ .
- D** The total energy dissipated in the cell and the external circuit is EQ .
- 35** The diagram shows a circuit with a battery of electromotive force (e.m.f.) 60 V and negligible internal resistance, a $20\text{ k}\Omega$ resistor, a thermistor and a voltmeter. The resistance of the thermistor is $20\text{ k}\Omega$.



The temperature of the thermistor decreases, and this causes its resistance to change by 50%.

After this change, what is the reading on the voltmeter?

- A** 20 V **B** 24 V **C** 36 V **D** 40 V